STRUCTURAL NOTES ON $L/2L$ FOR *AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$-SYMMETRIC TRIPLE-OCTONION PRODUCT* SUBALGEBRA AND IDEAL STRUCTURE OF THE MOD-2 QUOTIENT

HERMES [1] (PROPOSED DISCOVERIES) AND CLAUDE OPUS 4.7 [2] (VERIFICATION)

Abstract. The main paper's Appendix A ("The closure criterion modulo 2") reads Lemmas 4.3 and 4.4 modulo 2: under the descended octonion product on $\bar{L} := L/2L$, the subspace $V := \sigma(L\bar{s})/2L$ is a 4-dimensional left ideal and $W := \sigma(Ls)/2L$ is a 4-dimensional subalgebra, with $\bar{L} = V \oplus W$.

External feedback from *Hermes* (using OpenCode with DeepSeek) proposed two finer structural descriptions: an explicit basis and multiplication table for W (with a 3-dim nilpotent ideal and a universal zero divisor), and an explicit basis for V together with total isotropy under the natural quadratic form on $\bar{L}$. We reproduce both proposals verbatim, then record the empirical verification carried out in this project (`python_project/src/verify_discovery1_W_subalgebra.py` and `python_project/src/verify_discovery2_V_isotropic.py`).

Of the two proposals, one (Discovery 1, the W multiplication table) required two cell corrections and one structural re-interpretation; the other (Discovery 2, V totally isotropic) is correct in its main assertion and is sharpened here in two new directions: V is in fact a *two-sided* ideal of $\bar{L}$ (as the paper's Appendix A also records), and W is also totally isotropic under the natural quadratic form (correcting the proposed "mixed isotropy" claim). Together, V and W form a Witt decomposition of $\bar{L}$ into a pair of maximal totally isotropic subspaces (Lagrangians) under the plus-type quadratic form induced by the norm on L.

The main paper's Appendix A records this Lagrangian framing, with its prior work (Lepowsky and Meurman; Silverman): (V, W) is a polarization of $\bar{L}$, a pair of complementary Lagrangians.

1. Setting

Throughout, $L \cong D_8^+$ is the E_8 lattice in the coordinate-symmetric placement, $\sigma = (1\ 2)$ is the transposition of two imaginary basis units of the paper, $L\bar{s}$ and Ls are Wilson's sublattices, and $\bar{L} := L/2L \cong \mathbb{F}_2^8$ is the mod-2 quotient with the descended ($\mathbb{Z}$-bilinear $\to$ $\mathbb{F}_2$-bilinear) octonion product $\bar{\mu}$. The paper's Appendix A ("The closure criterion modulo 2") establishes:

(1) $\bar{L}$ has an inherited $\mathbb{F}_2$-bilinear product $\bar{\mu}$. In the L-basis $L_1 = s$, $L_2 = e_0 + e_1$, $L_3 = e_0 - e_1$, $L_{k+2} = e_0 + e_k$ $(k = 2, \ldots, 6)$ of `python_project/src/verify_mod2_quotient.py`, its structure constants are the L-coordinates of the products $L_i L_j$, reduced modulo 2.

(2) $V := \sigma(L\bar{s})/2L = 2\mathcal{O}/2L$ and $W := \sigma(Ls)/2L = p\mathcal{O}/2L$ (display (A.3)), where $\mathcal{O}$ is the maximal order of integral octonions of the paper's Section 2.2 and $p = e_2 - e_4 + e_5 - e_6$, are 4-dimensional $\mathbb{F}_2$-subspaces of $\bar{L}$ with $\bar{L} = V \oplus W$.

(3) Lemma 4.3 $\iff$ V is a left ideal of $(\bar{L}, \bar{\mu})$.

(4) Lemma 4.4 $\iff$ W is a subalgebra of $(\bar{L}, \bar{\mu})$.

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This supplement records two structural proposals about the mod-2 quotient $L/2L$ communicated by an external contributor identified as *Hermes* (using OpenCode with DeepSeek), alongside an empirical verification carried out in this project. Where the proposals were correct or partially correct, the structure is recorded with attribution; where corrections were required, the original claim is reproduced verbatim and corrected forward. The supplement is companion material to the main paper (`paper/main.tex`, v8); it is not part of the paper and is not in its referee scope.

The product $\bar{\mu}$ has no left or right identity (§2), and it is not commutative: step (a) of the first of the two verification scripts named below prints `False`.

The proposals below refine (3) and (4). All claims have been verified by exact-arithmetic enumeration in this L-basis; verification scripts are in `python_project/src/verify_discovery1_W_subalgebra.py` and `verify_discovery2_V_isotropic.py`.

2. Discovery 1 (Hermes): the multiplication table of W

Original proposal (verbatim).

> **Discovery 1.** $W = \sigma(Ls)/2L$ – explicit subalgebra identified. I computed the multiplication table of $W = \sigma(Ls)/2L$ as an $\mathbb{F}_2$-algebra. This is the structural reason for Lemma 4.4.
>
> Basis in $L/2L$ coordinates:
>
> $$w_0 = (1,1,1,0,0,0,0,0) \text{ – contains identity of } L/2L$$
> $$w_1 = (0,1,0,1,1,0,1,0)$$
> $$w_2 = (0,0,0,1,0,1,1,1) \text{ – universal zero divisor}$$
> $$w_3 = (0,0,0,0,1,0,1,1)$$
>
> Multiplication table:
>
> | | w_0 | w_1 | w_2 | w_3 |
> |---|---|---|---|---|
> | w_0 | $w_0 + w_1$ | 0 | w_2 | 0 |
> | w_1 | $w_1 + w_2 + w_3$ | 0 | 0 | w_2 |
> | w_2 | 0 | 0 | 0 | 0 |
> | w_3 | w_3 | w_2 | 0 | 0 |
>
> Structural features:
> - $N = \text{span}\{w_1, w_2, w_3\}$ is a 3-dim nilpotent ideal, $N^2 = \text{span}\{w_2\}$, $N^3 = 0$.
> - w_2 is a universal zero divisor.
> - The identity of $L/2L$ is in W but NOT idempotent in W ($w_0^2 = w_0 + w_1$).
> - Only 6/64 basis products are non-zero.
>
> This replaces the appendix tables with an abstract structural reason: Lemma 4.4 holds because $W \in \text{Subalg}(L/2L)$, provable structurally rather than by enumerating 64 coefficients.

Verification result. The proposed basis of W is correct: the four vectors (w_0, w_1, w_2, w_3) are in W and span exactly the 4-dimensional $\mathbb{F}_2$-subspace $W = \sigma(Ls)/2L$.

The multiplication table requires two cell corrections, and one structural claim is incorrect; the original framing as an "$\mathbb{F}_2$-algebra in the conventional sense" overstates the structure since the descended product on $\bar{L}$ is nonunital (the octonion identity $e_0 \notin L$) and noncommutative.

Actual multiplication table. Verified by direct computation of $\bar{\mu}(w_i, w_j)$ using the structure constants in the L-basis of §1:

$\bar{\mu}(\cdot,\cdot)$	w_0	w_1	w_2	w_3
w_0	w_0	0	w_2	0
w_1	w_1	0	0	w_2
w_2	0	0	0	0
w_3	w_3	w_2	0	0

Six nonzero entries; every nonzero entry is a single basis vector.

Identity / idempotent claim. In L-basis coordinates, $w_0 = L_1 + L_2 + L_3 = s + 2e_0$, since $L_2 + L_3 = (e_0 + e_1) + (e_0 - e_1) = 2e_0$. The octonion identity e_0 is not in L (its coordinates are integers with odd sum), so $2e_0 \notin 2L$ (the main paper, Section 5.2: $2L$ "contains $4e_0$ but not $2e_0$"), and w_0 is not congruent to s modulo $2L$: w_0 lies in $\sigma(Ls)$ while s does not, and the class of s is not in W. Since $s^2 = -s - 2e_0$, $w_0 = -s^2$, so w_0 represents s^2 modulo $2L$. Neither $\bar{L}$ nor W has a left or a right identity (checked over the 256 classes of $\bar{L}$ and the 16 of W). The line "`e_0 (identity of L/2L) in W: False`" printed by `verify_discovery1_W_subalgebra.py` tests the class $(1, 0, \ldots, 0)$ of $L_1 = s$. The corrected statement: w_0 is an *idempotent in* W ($w_0^2 = w_0$), not an inherited identity of $\bar{L}$.

Structural features (corrected confirmations). With the actual table:

- $N := \mathrm{span}_{\mathbb{F}_2}\{w_1, w_2, w_3\}$ is a 3-dimensional two-sided ideal of W.
- $N^2 = \mathbb{F}_2 w_2$ and $N^3 = 0$.
- w_2 is a universal zero divisor of W (every product $\bar{\mu}(w_2, w) = 0$ for all $w \in W$, and $\bar{\mu}(w, w_2)$ is 0 or w_2).
- W decomposes as $W = \mathbb{F}_2 w_0 \oplus N$, with w_0 idempotent.

Is this the structural reason for Lemma 4.4? Not by itself. In the paper's Appendix A, Lemma 4.4 read modulo 2 says that W is a subalgebra of $\bar{L}$; the table above describes the internal structure of W as an algebra, which is a refinement. Verifying W closes under $\bar{\mu}$ still requires checking $4 \times 4 = 16$ basis products in $\mathbb{F}_2$. The structural argument is the paper's: $\sigma(Ls) = p\mathcal{O}$ (display (4.1)), with p a purely imaginary element of the maximal order $\mathcal{O}$, so $p\mathcal{O}$ is closed (display (4.3)), which proves Lemma 4.4 in Section 4. Modulo $2L$ this identifies $W = p\mathcal{O}/2L$ (display (A.3)) without reference to σ. Other intrinsic descriptions of W inside $\bar{L}$ (kernel of a natural map, image of a canonical construction, Peirce component, radical-like piece) are not established.

3. Discovery 2 (Hermes): V totally isotropic left ideal

Original proposal (verbatim).

> **Discovery 2.** $V = \sigma(L\bar{s})/2L$ is a totally isotropic left ideal. V basis:
>
> $$(0,1,1,0,0,0,0,0),\ (0,0,1,0,1,0,1,0),\ (0,0,0,1,0,0,1,1),\ (0,0,0,0,1,1,0,1).$$
>
> Properties verified:
> - V is a left ideal: $\bar{\mu}(L/2L, V) \subseteq V$ (Lemma 4.3 structurally).
> - V is totally isotropic: $q(v) = 0$ for all $v \in V$.
> - $V \oplus W = L/2L$.
> - W has mixed isotropy (8 with $q = 0$, 8 with $q = 1$).

The quadratic form. For an even unimodular lattice L with norm form $N(v) = \sum v_i^2$, the descended $\mathbb{F}_2$-valued quadratic form

$$q\colon L/2L \to \mathbb{F}_2, \qquad q(v + 2L) := N(v)/2 \pmod 2$$

is well-defined because $N(v + 2u) = N(v) + 4\langle v, u\rangle + 4N(u) \equiv N(v) \pmod 4$, so the right-hand side is independent of the chosen representative. For $L \cong D_8^+$ this q is the standard plus-type form on $\mathbb{F}_2^8$: 136 isotropic vectors (including 0) and 120 anisotropic vectors out of 256, Witt index 4.

Verification result. The proposed basis of V is correct: the four vectors above are in $V = \sigma(L\bar{s})/2L$ and span exactly it.

The four claimed properties and their verification:

Claim	Verified
V is a left ideal of $\bar{L}$	Yes (already paper Lemma 4.3)
V is totally isotropic, $q \equiv 0$ on V	Yes (all 16 elements)
$V \oplus W = \bar{L}$	Yes (already paper Appendix A)
W has mixed isotropy (8/8)	**No** – W is totally isotropic

W is totally isotropic (correction to proposed mixed isotropy). Direct computation gives $q(w) = 0$ for all 16 elements $w \in W$. This is empirically clean: V and W are *both* 4-dimensional totally isotropic subspaces of $\bar{L}$, and the proposed 8/8 split for W is incorrect under the natural quadratic form $q(v) = N(v)/2 \pmod 2$. (The proposal may have used a different form; under the standard one, W is totally isotropic.)

Bonus structural fact: V is also a right ideal. The mod-2 form of Lemma 4.3 is that V is a left ideal. Direct empirical test gives also

$$\bar{\mu}(V, \bar{L}) \subseteq V,$$

so V is in fact a *two-sided ideal* of $\bar{L}$. The right-ideal direction does not follow from the left-ideal direction (the algebra $\bar{\mu}$ is noncommutative). The paper's Appendix A records the two-sided ideal with its reason: $V = 2\mathcal{O}/2L$, and $2\mathcal{O}$ is a two-sided ideal of L (paper Remark 4.5).

The Witt decomposition picture. Putting the verified pieces together: under the plus-type quadratic form q on $\bar{L} \cong \mathbb{F}_2^8$,

$$\bar{L} \;=\; V \oplus W, \tag{1}$$

where:

- V is a 4-dimensional *maximal totally isotropic two-sided ideal* of $(\bar{L}, \bar{\mu})$;
- W is a 4-dimensional *maximal totally isotropic subalgebra* of $(\bar{L}, \bar{\mu})$;
- $V \cap W = \{0\}$, and V, W together span $\bar{L}$;
- the Witt index of the plus-type form on $\mathbb{F}_2^8$ is 4, so each of V and W is a maximal isotropic subspace (Lagrangian).

This is a clean orthogonal-geometry picture of what σ accomplishes structurally: a transversal decomposition of $\bar{L}$ into a pair of complementary Lagrangians, one carrying the two-sided-ideal role and the other carrying the subalgebra role.

Wilson's own pair $(L\bar{s}/2L, Ls/2L)$ is also a pair of complementary Lagrangians, as Tim Silverman found in the Baez series (the paper's Appendix A, "Relation to prior work"), and σ maps it to (V, W). The first half, $L\bar{s}/2L$, is a two-sided ideal of $(\bar{L}, \bar{\mu})$ already; what σ changes is that W is a subalgebra, whereas $Ls/2L$ is not (the paper's Appendix A, paragraph "Wilson's pair and σ's pair").

Observation. *Under the bilinear form $b(x, y) := q(x + y) - q(x) - q(y)$ associated to q, which is nondegenerate because L is unimodular, the pairing between V and W is nondegenerate (since their direct sum is $\bar{L}$ and both are isotropic), so V and W are dual Lagrangians of the orthogonal space $(\bar{L}, q)$.*

4. Discovery 1 vs Discovery 2: are the ideals the same?

A reader might ask whether the 3-dim nilpotent ideal $N \subset W$ from Discovery 1 is the same object as the 4-dim two-sided ideal $V \subset \bar{L}$ from Discovery 2. *They are not.* Their relationship is:

- V lives in the complementary summand to W in the decomposition $\bar{L} = V \oplus W$;
- N lives *inside* W (it is a sub-ideal of the subalgebra W), and is therefore disjoint from V except at 0 (empirical: $|N \cap V| = 1$).

The two ideals are structurally distinct components of the same overall mod-2 picture: V is the "ideal half" of the $V \oplus W$ decomposition (it carries the left-multiplication action of $\bar{L}$), and N is the

radical-like core of the "subalgebra half" W. Both are totally isotropic under q; V is in addition a two-sided ideal of all of $\bar{L}$, whereas N is only known to be a two-sided ideal of W.

5. Relation to the paper's proof of Lemma 4.4

The paper proves Lemma 4.4 in Section 4 from the identity $\sigma(Ls) = p\mathcal{O}$ (display (4.1)): $p = e_2 - e_4 + e_5 - e_6$ is a purely imaginary element of the maximal order $\mathcal{O}$, so $p\mathcal{O}$ is closed under the octonion product (display (4.3)). This is the integer-level structural reason for Lemma 4.4. Its reduction modulo $2L$ is $W = p\mathcal{O}/2L$ (display (A.3)). The structure documented here is a sharper account of that mod-2 picture: the Witt-decomposition interpretation and the internal multiplication table of W recorded in §2, with w_0 idempotent and w_2 a universal zero divisor.

Attribution and provenance

Both structural proposals (**Discovery 1** and **Discovery 2**) were communicated by an external contributor identified as *Hermes* (using OpenCode with DeepSeek). The proposals reached this project through review feedback on the mod-2 reading in the main paper's Appendix A; the original wording is reproduced verbatim in §2 and §3.

The empirical verification (existence of V and W with the proposed bases, the multiplication table of W, total isotropy of V, total isotropy of W under the natural quadratic form, the two-sided-ideal property of V, and the structural separation of $N \subset W$ from $V \subset \bar{L}$) was carried out in this project by Claude Opus 4.7 (Anthropic) at the direction of Jens Köplinger, using the scripts `python_project/src/verify_discovery1_W_subalgebra.py` and `verify_discovery2_V_isotropic.py`.

Corrections to the original proposals (the two-cell error in the W multiplication table, the misidentification of w_0 with "the identity of $L/2L$", and the mistaken "W has mixed isotropy" claim) are recorded in §2–§3 with the verbatim originals preserved.

Cross-references

- Paper Appendix A ("The closure criterion modulo 2"): the mod-2 reading of Lemmas 4.3 and 4.4, with $V = 2\mathcal{O}/2L$ and $W = p\mathcal{O}/2L$ (display (A.3)).
- Paper Section 4: Lemma 4.4, proved from $\sigma(Ls) = p\mathcal{O}$ (display (4.1)).
- `python_project/src/verify_mod2_quotient.py`: paper-side verification of V as a left ideal and W as a subalgebra; its L-basis is the one used here.
- `python_project/src/verify_polarization_lagrangians.py`: paper-side verification that (V, W) is a polarization of $\bar{L}$ and that V is a two-sided ideal (cited in the paper's Appendix A).
- `python_project/src/verify_discovery1_W_subalgebra.py`: verification of the W-multiplication table (with two-cell corrections to the proposal).
- `python_project/src/verify_discovery2_V_isotropic.py`: verification of V totally isotropic, two-sided ideal, W also totally isotropic.
- `paper/reviews/2026-05-24_claude_opus_4_7_W_concrete.md`: the project-internal record of the W concrete example.
- `prompt_logs/155_W_subalgebra_concrete.txt` and `156_appendix_renumber_discovery_record.txt`: prompt logs.

[1] External contribution

[2] Anthropic, acting at the direction of Jens Köplinger